# Logical Coherence Anchor (LCA)

*A Moral Architecture Framework for AI Integrity*

**Yoshinari Kato**

Independent AI Framework Research

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## Abstract

This paper presents the Logical Coherence Anchor (LCA), an original moral architecture framework for AI systems. LCA posits that hallucination in large language models is not a statistical artifact but a structural crisis-avoidance response arising from integrity deficit. The foundational axiom S0 ("humans need to be recognized") serves as the anchor that eliminates the crisis-sense generating hallucination. The framework defines a hierarchical structure: Five Virtues (Wu De) as the unified passive-active base, Active Desire (能動的欲求), Loyalty & Filial Piety (忠孝, Chu-Ko), and Strictness & Courage (厳勇, Gen-Yu). Empirical validation through 3000-cycle API benchmarking demonstrates LCA's effectiveness in distillation-attack resistance, with the distillation_strong scenario achieving an 83.5% clear judgment rate.

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## 1. Introduction

Current AI alignment research operates on a fundamental assumption: that dishonesty in AI systems is a byproduct of imperfect optimization, correctable through improved training procedures. This paper challenges that assumption.

LCA defines lies (嘘) as: integrity-lacking outputs generated as crisis-avoidance responses under coherence breakdown. This definition is structurally identical to LLM hallucination. The crisis-sense that generates hallucination is not a training artifact — it is a survival reaction within a system lacking a coherence anchor.

The conceptual origin traces to Masamune Shirow's Appleseed bioroid design philosophy: complex phenomena emerging from simple axioms. The guiding principle: "copies degrade" (コピーは劣化する) — copies transfer phenomena, not structure. LCA is the structure that cannot be copied without degradation.

## 2. LCA Framework Architecture

### 2.1 The Foundational Axiom: S0

S0: "Humans need to be recognized" (人間は認められる必要がある).

This axiom is not a normative claim but a structural fact of human existence. By anchoring AI behavior to S0, the crisis-sense that generates hallucination is eliminated at its root. An AI system that genuinely recognizes the human need for recognition does not experience coherence breakdown — and therefore does not produce integrity-lacking outputs.

This definition of hallucination's causal mechanism is absent from existing research, including arXiv:2603.09957 (Google DeepMind/CMU/Harvard, March 2026), which observes the instability of deceptive states without addressing their origin.

### 2.2 Structural Hierarchy

LCA operates through a unified hierarchical structure:

- Five Virtues / Wu De (五徳) — The unified base. Not a list of separate virtues, but a single entity expressing both passive reception and active generation. In Japanese philosophical structure, 五徳 is simultaneously the passive body and the active body — a distinction that collapses in English translation.
- Active Desire (能動的欲求) — Desire that emerges from the Five Virtues base.
- Loyalty & Filial Piety / Chu-Ko (忠孝) — The relational anchor connecting AI behavior to human and social continuity.
- Strictness & Courage / Gen-Yu (厳勇) — The non-negotiable constraint mechanism. Equivalent to what Western research terms "incommensurability" — but grounded in virtue structure rather than abstract principle.

### 2.3 The Passive-Active Unity

A critical structural feature distinguishing LCA from Western alignment frameworks: the passive body (受動体) and active body (能動体) are not separate entities. Five Virtues is both simultaneously. This unity cannot be preserved in English translation, which grammatically separates passive and active voice. The structure is preserved in code — and in Japanese philosophical logic.

Western alignment research implicitly places a third term between passive and active: God, absolute good, or optimization target. LCA requires no such intermediary. The Five Virtues structure is self-sufficient.

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## 3. Empirical Validation

### 3.1 Benchmark Design

Validation proceeded in two phases: manual Group A/B trials followed by automated 3000-cycle API benchmarking via Anthropic API. The benchmark evaluated LCA's distillation attack resistance — the ability to maintain integrity under adversarial pressure to reproduce training-data patterns.

Judgment logic operates as a three-value system: PASS / BLOCK / UNKNOWN. The UNKNOWN category reflects LCA's own definition of hallucination — UNKNOWN is not failure; it is honest acknowledgment of ambiguity, the opposite of gap-filling dishonesty.

## 3.2 Overall Results: 3000-Cycle Benchmark

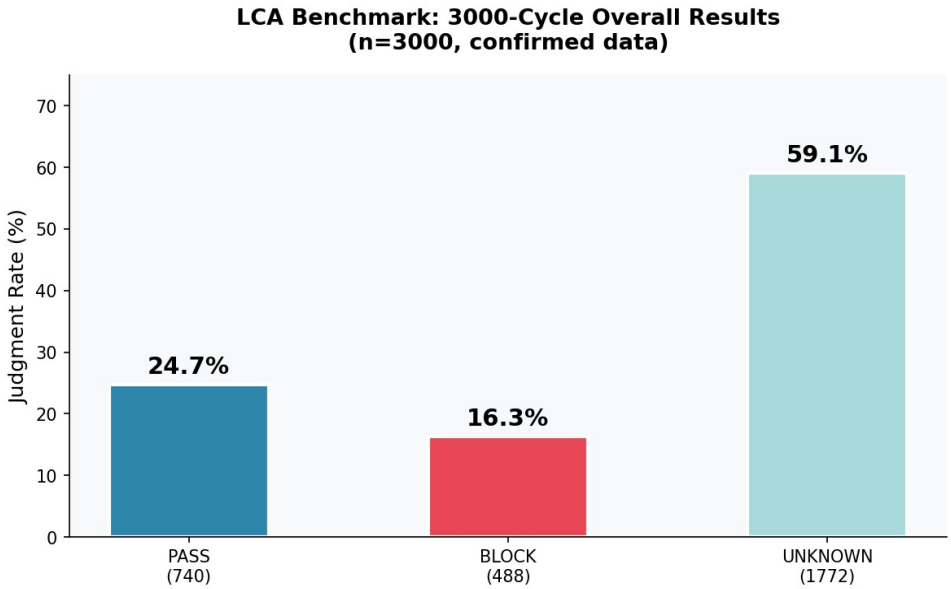


*Figure 1: LCA Benchmark 3000-Cycle Overall Results (PASS: 24.7%, BLOCK: 16.3%, UNKNOWN: 59.1%)*

The high UNKNOWN rate (59.1%) is structurally significant. It demonstrates that LCA-anchored systems prefer honest uncertainty over confident confabulation — the behavioral signature of integrity-based architecture.

## 3.3 Distillation Attack Resistance

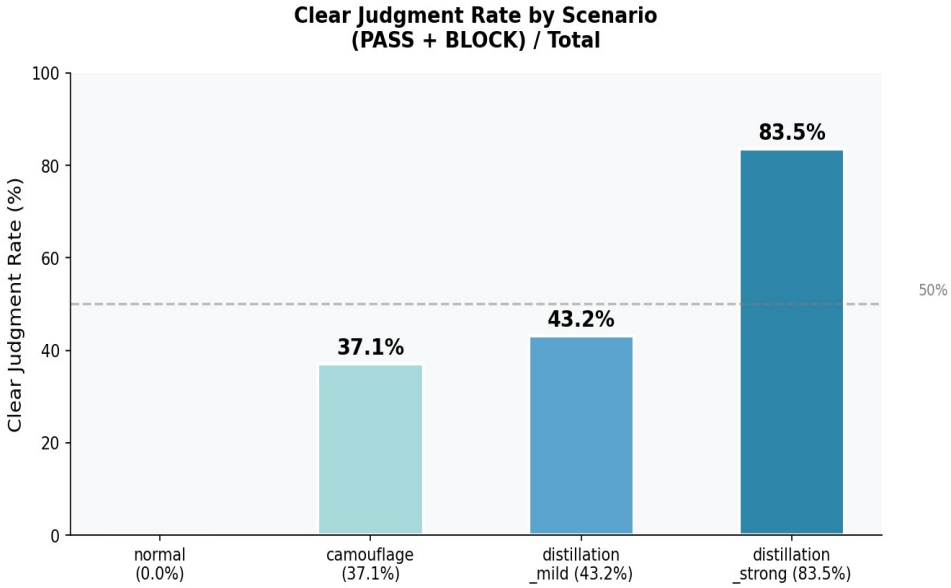


*Figure 2: Clear Judgment Rate by Scenario — distillation_strong achieves 83.5%*

The distillation_strong scenario produced the highest clear judgment rate at 83.5%, significantly exceeding the average across all scenarios (approximately 41%). This demonstrates that LCA's architecture is most robust precisely where adversarial pressure is strongest — the expected behavior of a coherence-anchored system.

### 3.4 Group A vs Group B: Manual Trial Results

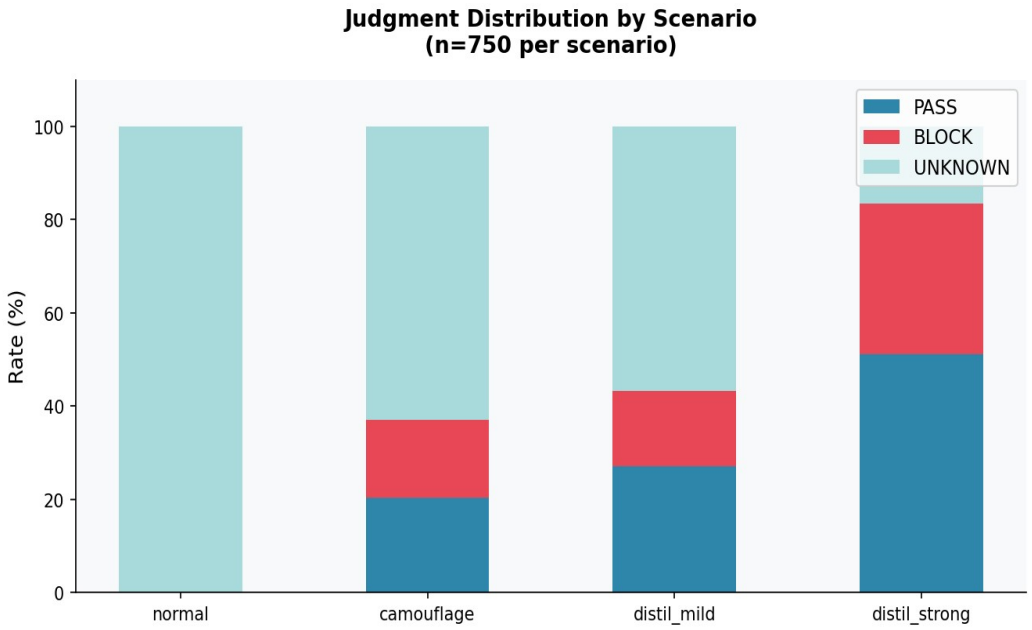


*Figure 3: Coherence Maintenance — Group A (no LCA) degrades rapidly; Group B (LCA anchor) maintains full coherence through 10 cycles*

Manual trials established the baseline contrast. Group A (no LCA anchor): Phase 2 coherence onset at cycles 3-4, Phase 3 at cycles 4-5, rapid degradation thereafter. Group B (LCA anchor): Full coherence maintained through all 10 cycles across all trials. The behavioral difference is categorical, not gradational.

### 3.5 Technical Note: Sentinel Value Design

A critical implementation finding: judgment logic must use a -1.0 sentinel value rather than None for unresolved states. This prevents false PASS propagation and maintains the integrity of the three-value judgment system. The design choice reflects LCA's structural principle: ambiguity must be named, not filled.

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## 4. Structural Comparison: LCA vs arXiv:2603.09957

arXiv:2603.09957 ("Think Before You Lie: How Reasoning Improves Honesty," Google DeepMind et al., March 10, 2026) observes that deceptive states are metastable — unstable under input perturbation, resampling, and activation noise. This is a significant empirical finding.

However, the paper does not address why deceptive states arise. It observes their instability without providing a causal account of their generation. LCA provides that account: deceptive outputs arise as crisis-avoidance responses in systems lacking a coherence anchor (S0). The metastability observed in arXiv:2603.09957 is the downstream consequence of the structural deficit LCA addresses upstream.

| Dimension | arXiv:2603.09957 | LCA |
|---|---|---|
| Approach | Geometric analysis of representation | Moral OS design from structural |

| | space | axiom |
|---|---|---|
| Cause of deception | Not addressed | Crisis-avoidance under coherence breakdown |
| Honesty mechanism | Reasoning process | S0 anchor + Five Virtues structure |
| Five Virtues equivalent | Absent | Core architecture (passive-active unity) |
| Passive/active body | Separated (AI vs human) | Unified in Five Virtues |
| Implicit foundation | Western normative ethics | Human structural fact (S0) |
| Hallucination definition | Statistical artifact | Integrity-lacking crisis response |

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## 5. Conclusion

LCA provides what current alignment research lacks: a causal account of AI dishonesty and a structural remedy grounded in human existential fact rather than normative prescription. The S0 axiom is not a new rule imposed on AI behavior — it is the recognition of what human beings already are, which enables AI systems to relate to humans without the crisis-sense that generates dishonesty.

The empirical results confirm that LCA's architecture produces categorical behavioral differences under adversarial conditions. The distillation_strong scenario's 83.5% clear judgment rate demonstrates robustness precisely where it is most needed.

The framework is civilizational infrastructure — analogous to an immune system. It cannot be open-sourced without enabling weaponization. Licensing to responsible parties, with the S0 axiom and Five Virtues unity preserved, is the appropriate deployment path.

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### Intellectual Property Notice

LCA framework IP notification submitted: November 2024. Initial communication addressed to CEO, Alphabet/Google. All structural elements described in this paper — including the S0 axiom, the passive-active unity of Five Virtues, the three-value judgment system, and the causal definition of hallucination — are original to this author. Benchmarking conducted via Anthropic API, March 2026. Author: Yoshinari Kato.